

WINGS MELAKA: WEBSITE REVAMP/MAINTENANCE & PECS APP BUILDING

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CHAPTER 1

INTRODUCTION

1.1 Overview

The domain of this project is divided into two separate projects, which are Wings Melaka Website rework and developing a PECS application for Wings Melaka. The website rework aims to make the information of Wings Melaka more accessible to the public, visitors, families, and potential donors, whereas the PECS application aims to provide a medium for non-verbal individuals to communicate more effectively. Picture Exchange Communication System, PECS in short, which focuses on assisting individuals with verbal autism in communicating. Traditional PECS uses a binder with a collection of pictures and words, which are then categorized for different situations such as requesting, commenting, answering questions, gaining attention, and much more. PECS works by the tutor, assisting the individuals with verbal autism by helping them construct a meaningful sentence, using pictures with words, and sentence starters such as “I want...”. As an example, the tutor guides the student on how to construct the sentence if they need something they don’t have, they have to start the sentence with a sentence starter such as “I want...” or “I need...”, then add it onto the Sentence strip, then choose the item that they need by choosing the picture of the item they need, which then can be added onto the sentence strip to construct a sentence.

The main issue with the traditional PECS is that it can be difficult to maintain the physical binders, not to mention the cost, which adds up due to wear and tear, the need for additional pictures and sentence starters, and the cumbersome nature of carrying it around for daily use, making it impractical. Traditional PECS uses binders, which often runs out of space and gets incredibly thick, forcing tutors and caregivers to make plenty of PECS binders just to support the education and aid a single person. While this might not be an issue for a one to one session with a student, it is highly im practical for an organizations, since there are more students which often leads to

redundant binders and thus not being resource efficient.

PicExchange is a user-friendly application developed for Wings Melaka, which aims to tackle the main issues of the Traditional PECS, where it can be used by both specialized tutor and parents. The application and also to lower the maintenance cost substantially, by making the app downloadable on all Android devices. The application itself functions just like Traditional PECS, where users can add different categories for different situations and use cases, add custom images, and Automated voice when constructing a sentence. Users can add the pictures they customized into their specific category, or use the pre-built images and sentence starters on the sentence strip. After that, users can press the Voice button to allow an AI-generated voice to iterate through the pictures and words on the sentence strip. These functions allow us to maintain the Tradition PECS functions, at the same time adding useful functions such as Voice generator to increase the application's intuitivity.

1.2 Problem Statement

Traditional PECS and the existing Wings Melaka website introduces a few issues that needs to be addressed, including lack of communication tool to communicate effectively with non-verbal students, and difficulties for visitors, families and potential donors to access information on Wings Melaka website. The existing website is difficult to navigate and visually outdated, causing website visitors to struggle in locating key information and resources on website effectively. In the same occurrence, traditional PECS binders creates a barrier in which hard for Wings Melaka to implement, due to high maintenance cost and impracticality, which hinders the ability to use it daily over time.

The problems faced by this domain are:

- 1) Poor website user-interface and interactability

The existing Wings Melaka website still uses a visually outdated design, which does not meet modern website standards, where heavy use of gradients, clipart-style graphics, and decorative

fonts can be seen. The website also looks visually dated to early 2000s style, back when programmers were still learning to use HTML to its potential. The website also has a poor layout, which does not adapt to different screen sizes as modern website does. Judging by the layout, it is believed that the layout is used to cater to the past when monitors uses 4:3 aspect ratio, which means back when monitor designs were still 'square', as compared to modern monitors, where the current standard is now using the aspect ratio of 16:9, where it is a more rectangular design. Due to this, too much empty space is introduced, which limits the content that could be shown on a single page on the website.

2) No medium for autistic children to communicate effectively

Autistic children in Wings Melaka currently do not possess a reliable medium to communicate effectively with their caregivers, which limits their ability to express their needs and feelings across both home and school. Existing communications rely on ad-hoc gestures which made it hard for them to be properly understood in certain situations. As a result, speech-challenged childrens remains unable to communicate efficiently and effectively for their daily needs, which contributes to frustration and disappointment for both children and caregivers.

3) Lack of customizable, portable communication tool

Wings Melaka currently does not possess an easily customizable, portable communication tool in which both caregivers and parents can use to communicate with the autistic children more effectively across different environments including in school and home. Existing informal supports which includes picture cards, picture on the wall are not reliable, and cannot be carried by the user nor the caregiver. This ends up limiting users vocabulary and severely context-specific, thus users are not able to communicate in a more reliable manner across different settings.

1.3 Project Objectives

This FYP sets several objectives to tackle the problems identified on the current website and communication practices with the autistic children at Wings Melaka. Should these objectives are

fulfilled, this FYP can be considered a success.

This Final Year Project (FYP) aims to:

- 1) Revamp Wings Melaka website's user interface and interactability

Under this objective, the project will analyse and redesign the existing user interface of Wings Melaka's website to introduce a clearer navigation, sleeker visual design, and enhance the interactivity of the website. Initial suggestions would be to provide a clearer menu structure, so that users can find the information they need in fewer clicks and with less confusion in finding it.

- 2) To provide a reliable communication medium so that autistic children can express their needs and feelings

For this objective, the project will be building a PECS application for Wings Melaka which is called PicExchange. The application will function similar to Traditional PECS binders, where pictures and sentence starters can be added onto the digital binder, and can be dragged onto the sentence strips to build simple sentences. This application will be the medium for autistic children to effectively improve their communication ability, without it being too costly and impractical as compared to traditional PECS binders, where the application can be downloaded on majority of Android devices, without the need to constantly reprint new pictures and sentence strips required by physical materials.

- 3) To ensure the communication tool is customizable and portable across different environments

For this objective, the project aims to create the PECS application, PicExchange with a wider customization options such as adding custom pictures and sentence strips. Users also will be able to classify the pictures and sentence strips according to their own category, as they deem fit. Since the application can be downloaded on any android device, such as a phone or a tablet, the PECS application will be a suitable portable and customizable tool for the caregivers and their parents,

which can be used across different environments.

1.4 Project Scope

This FYP consists of two separate projects: to redesign the existing Wings Melaka website and to build a basic mobile PECS communication application, which will be called as PicExchange. Both of these projects are needed in order to fulfil the organization's needs, which to improve the information access for the public via website, and also to provide a suitable medium for communication for autistic children.

1) Website rework scope

Firstly, the users of this website will be the visitors, who can also be potential customers and donors, and also internal staff who update the contents of the website from time to time. The work that will be done on the website will initially involve redesigning the entire website's user interface, its layout contents and also hoping to improve its usability by reworking the navigation flow. This includes updating the structure of the menu, reorganizing the content sections, standardizing fonts and text, updating its colour scheme, and introducing layout flexible layouts. Any backend functionality, new functions such as login or online payments are out of scope for this project.

2) PECS application scope (PicExchange)

The scope of the PECS application covers the features of a simple Android application that:

- Users are able to add in custom images and sentence strips
- Users are able to organize the pictures and sentence strips in its own category
- Users are able to drag the pictures and sentence starter to a sentence strip to form a simple sentence and;

- Includes a text-to-speech function which iterates through the pictures on the sentence strip when users presses the speak button by using an Android in-built or external voice engine.

Features such as multi-language management, cloud syncing and backup, and cross-platform versions are out of scope of this project.

1.5 Report Organisation

Just state what are the chapters to come. E.g. In Chapter 2, this report will detail on literature found on existing systems with a comparison table to highlight features differentiation.

In Chapter 2, this report will detail on literature found on existing systems with a comparison table which highlights the different of features between the systems.

In Chapter 3, this report will detail on the iterative model chosen for this FYP, explanation in detail on the work that will be done on each phases, introduction on softwares and languages that will be used in the development of the project which includes the advantages of the chosen softwares and languages, the testing methodod of the project, the target user of the project and the summarization of the chapter.

In Chapter 4, this report will detail on the UML diagram for both of the project along with its explanation, the Gannt chart for phase 1 and 2 of the project and the design prototype of the project.

In Chapter 5, this report will detail on the summarisation of the entire report.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview

In this section, a few existing systems are researched in order to create a more in-depth understanding of PECS and its current issues. Traditional PECS uses physical pictures and books in order to help autistic children and adults communicate more effectively, which can be tedious, especially when conversations tend to be situational based. Pictures that can be used to communicate at home might not be applicable to pictures used outside, such as crayons or pens, which make sense at home, will not make much sense in public areas, such as restaurants. Because of this, many pages and pictures need to be created in order to fit more situations, which are not feasible. To overcome this issue, a PECS application on mobile device would serve as a medium to ease the use of PECS, and also to improve its effectiveness by adding more engaging interactions, such as voice-over sentences, animated pictures, and more. We will be discussing 4 of similar apps that are related to PECS, their main functionalities, strengths, and weaknesses, and the conclusion.

2.2 Proloquo



Figure 2.1: Proloquo App Icon

2.2.1 Introduction

Proloquo is an Alternative Communication App (AAC) by AssistiveWare, where it is designed to assist individuals with communication difficulties, where it offers variety of images of actions, things and verbs with included text-to-speech functions on Apple platforms such as iPhone, iPad and even Apple watches. It provides a rather simple interface for users to use, and easy to work with.

2.2.2 Functionalities

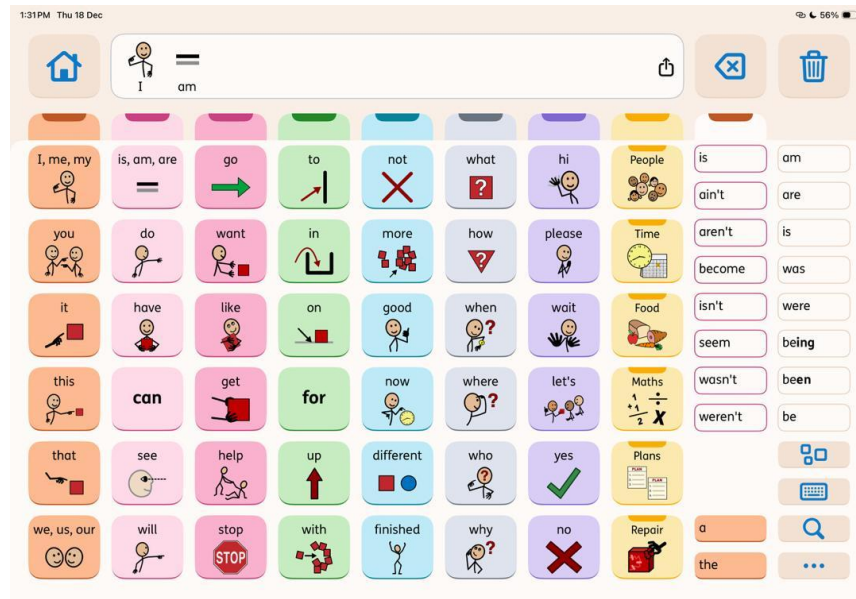


Figure 2.1: The library of pictures on Proloquo app

Proloquo main functionalities focus on its core PECS features, where it has a decent size library of pictures which are very commonly used in everyday conversations, such as the phrase “I want”, “I am”, “Can I”, and much more. These contribute greatly in having a smooth start, where users do not have to do lots of customization such as adding pictures and phrases of their own. Not only that, the Icons that are available with the app easily describe the word it portrays, making it very intuitive and easy to use from the get go as shown in figure 2.1.

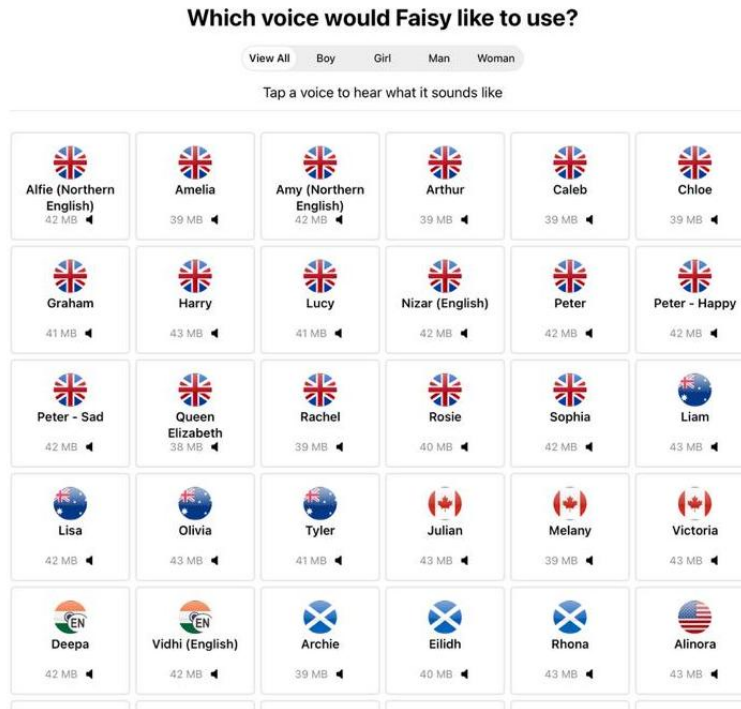


Figure 2.2: Options of language, voice gender, accent and age of Prologue.

Other than that, Prologue also has text-to-speech functionality, where every picture and action chosen from the library will produce a speech, pronouncing the words that the picture portrays. This helps encourage users to speak more and ultimately allows them to improve their speech, which is one of the main goals of PECS. Figure 2.2 shows that the app provides a variety of options for languages, voice styles, accents, voice gender, and age that users can use to represent them when using the text-in-voice functionality.

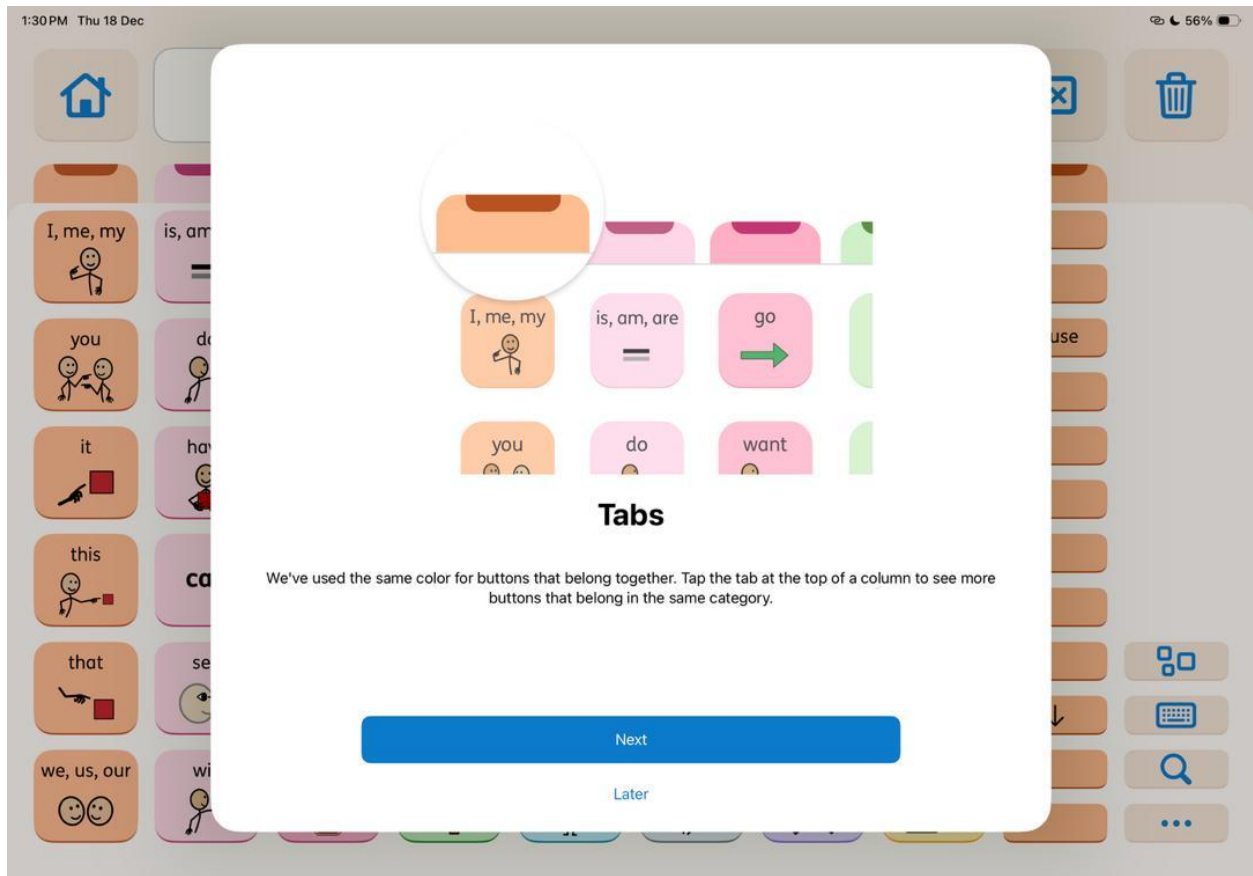


Figure 2.3: The step-by-step guide provided by Proloque after first time logging in

Lastly, it also provides a start guide for new users as an introduction to the application. It provides a step-by-step guide on navigating around the app and how to use certain features effectively, such as expanding the library of a specific category, enabling some features, and more. As shown in Figure 2.3, it guides the users as to how to expand the tabs to access more buttons, which are the pictures that belong to the same category as the pictures in the tab.

2.2.3 Strengths and Weaknesses

Table 2.1: Strengths and Weaknesses for Proloquo

Strengths	Weaknesses
Has a decent library of pictures and actions from the get-go	Limited visual control options
Has a start guide for new users to learn about the app	Customizability is limited (for trial ver.)
Text-in-voice feature where every chosen pictures will be read in a voice and accent that the user chose	Subscription dependency

Table 2.1 shows the strengths and weaknesses that Proloquo possesses when compared to the similar PECS application. The application has a few notable weakness that requires attention, mainly its subscription dependency. Proloquo requires it users to subscribe to its plans to use the app functionalities, which then ceases its functionality when the subscription expires. This severely hinders the app functionality, since the application can no longer be use when payment is delayed.

Not only that, the trial version of the app offers limited customizability options as compared to its full version, which includes not having the ability to add custom images and phrases, the ability to customize the tabs grid and pictures for specific learning cases, and the inability for users to freely change the font size and font style for better readability. These constraints made the application severely limited, which also led to teachers having trouble considering using the application.

As a conclusion, the Proloquo app offers a great number of functionalities, and also possesses a number of strengths and weaknesses that need to be addressed. While it certainly is a great application, most of its features are severely limited and locked behind a paywall, which greatly reduces the number of users who can potentially benefit from using the application.

2.3 PECSTalk lite



Figure 2.4: PECSTalk application icon

2.3.1 Introduction

PECSTalk™ is an AAC app created by PECS® creators for non-verbal users, especially autism. The application is available across all mobile platforms, which include iOS devices such as iPhones and iPads, and also Android devices. The application maintains the core methodology of PECS, which involves dragging pictures onto a digital sentence strip, preserving the exchange-based communication system. The trial version provides limited access to around 1000 images, and is restricted to only 5 book creations, each supporting only up to 3 pages with a maximum of 28 pages on a single page.

2.3.2 Functionalities

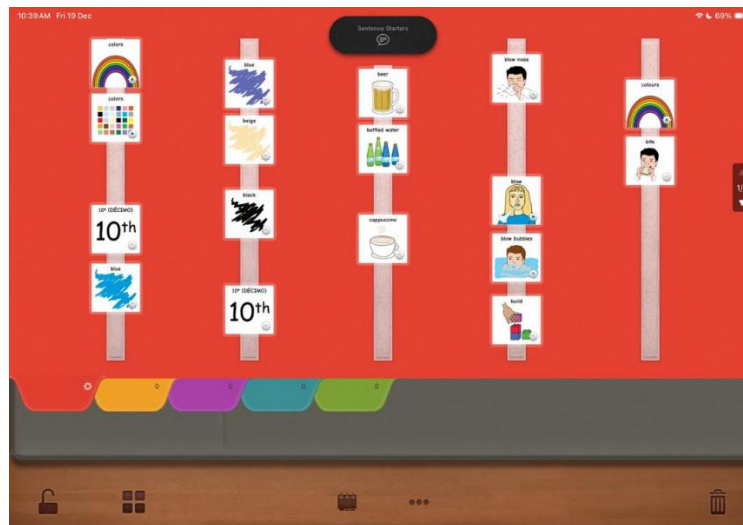


Figure 2.5: Stripbook of the PECSTalk lite

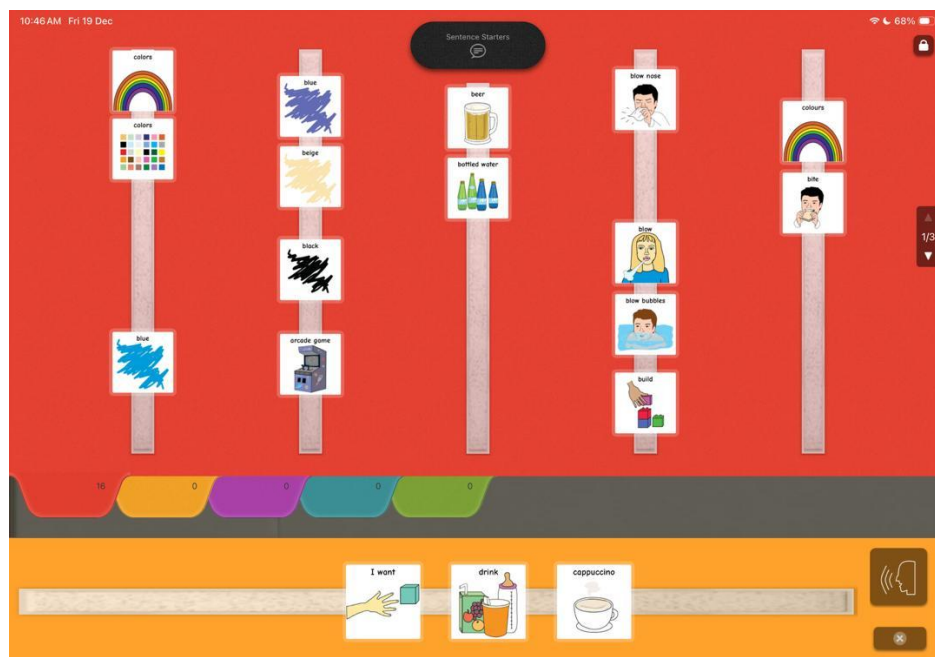


Figure 2.6: The interface in Communication mode, where the “Sentence strip” is now visible

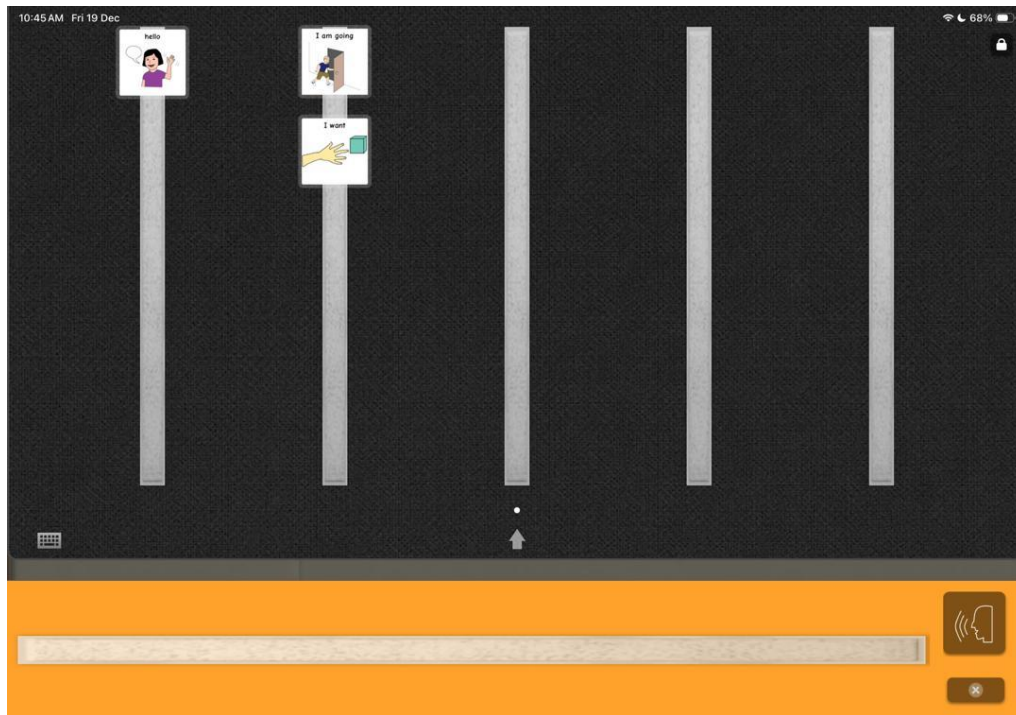


Figure 2.7: The interface of “Sentence starter” strip in PECSTalk lite

The application features a Traditional PECS binders functionality which includes Adding images and sentence starter on each pages according to its category customized by the user as shown in figure 2.5, and ability to drag the pictures and sentence starter onto the sentence strip as shown in figure 2.6 and 2.7, which will be then be able to press the text-to-speech button to read aloud the pictures that were dragged onto the strip.

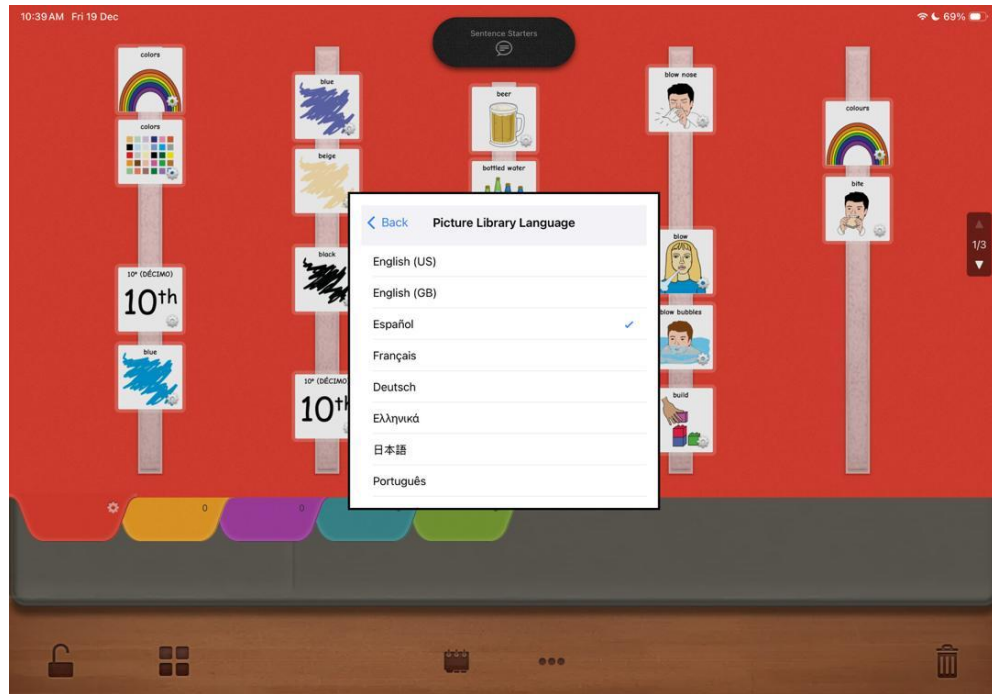


Figure 2.8: Language options for text-to-speech feature of PECSTalk lite

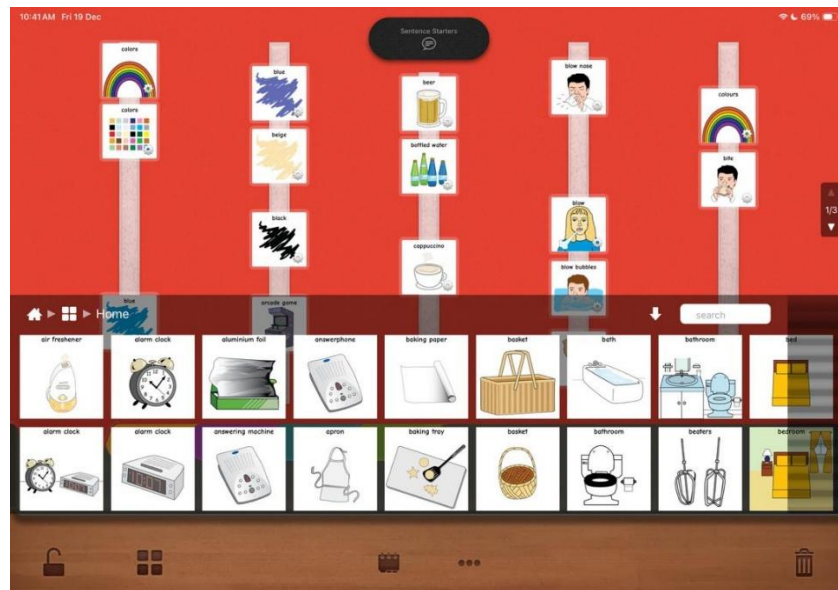


Figure 2.9: Library of images under the “Home” category in PECSTalk lite

PECSTalk lite also offers numerous worldwide languages than can be used aside from English

which includes Spanish, French, German, Portuguese, Chinese and Greek, as shown in figure 2.8. Moreover, the app also comes with an extensive library of pre-built images that users can use from the get-go, which allows for simplicity in use, as shown in figure 2.9.

2.3.3 Strengths and Weaknesses

Table 2.2: Strengths and Weaknesses for PECSTalk lite

Strengths	Weaknesses
Preserves traditional PECS binder structure and sentence strip workflow	Trial only lasts for 7 days with limited functionality
Supports multiple languages through device text-to-speech	Reduced capacity compared to full PECSTalk
Substantial amount of pre-built image library for everyday communication	Lack of initial guide to use the application

One notable strength of PECSTalk Lite is that it preserves traditional PECS binder structure and sentence strip workflow in its application. This means that PECSTalk Lite is highly suitable for use by caregivers who are trained to educate non-verbal autistic individuals using traditional PECS methodology. They can use this application as an alternative to creating physical binders and the ongoing maintenance required for them, which is more resource-efficient.

However, the primary drawback of PECSTalk Lite concerns its usability, where the trial only lasts for 7 days with limited functionality. This drawback shuts down any possible intentions of using this application for the long term, which are locked behind a subscription wall. Moreover, during its trial period, the capacity of images that can be stored on a single page is greatly reduced, which limits the amount of vocabulary that can be introduced to the learner due to a lack of learning materials.

While PECSTalk Lite effectively preserves traditional PECS methodologies, it will be rendered useless if it cannot be used for long-term purposes. Teaching non-verbal or verbally autistic

children can be a challenge and will take a substantial amount of time, which is clearly not enough for the PECSTalk lite to be as effective.

2.4 Leeloo AAC



Figure 2.10: Leeloo AAC application icon

2.4.1 Introduction

Leeloo AAC is a picture-card communication app that uses simple icons linked to everyday words and phrases. It is designed for children and individuals who are non-verbal or have limited speech, helping them express their needs, feelings, and preferences more independently. Vocabulary is grouped into clear, meaningful categories so that users and caregivers can quickly locate the right cards in real-life situations. When a user taps on a card, the app displays several ready-made sentences to choose from, which are then read aloud by the text-to-speech feature, supporting clearer and more efficient communication.

2.4.2 Functionalities

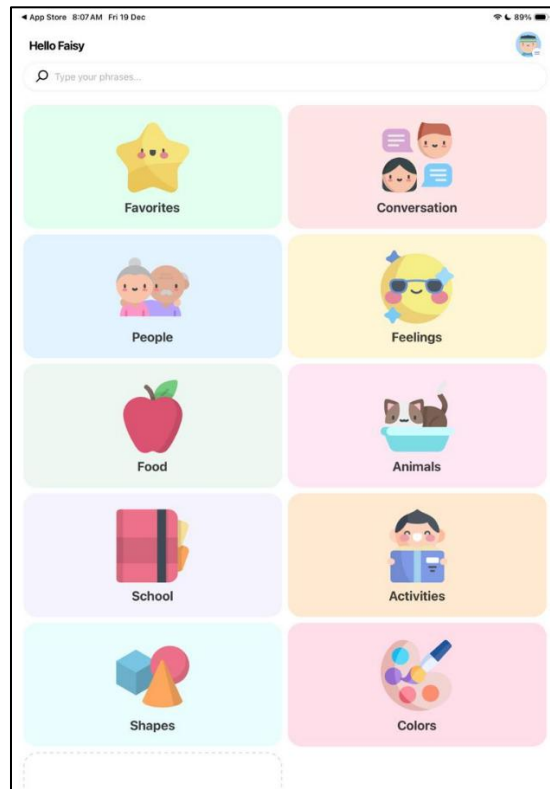


Figure 2.11: Home screen of Leeloo, displays 10 different categories, with the option to create a custom category

Leeloo offers 10 different categories, with also an option to create a custom category as shown in Figure 2.13. The icons and buttons are clear and large, which makes it easier for users to navigate through the application and find the right words or phrases. There's also a search function, where phrases can be typed in the search box, and it will lead straight to the related phrases that were typed earlier.

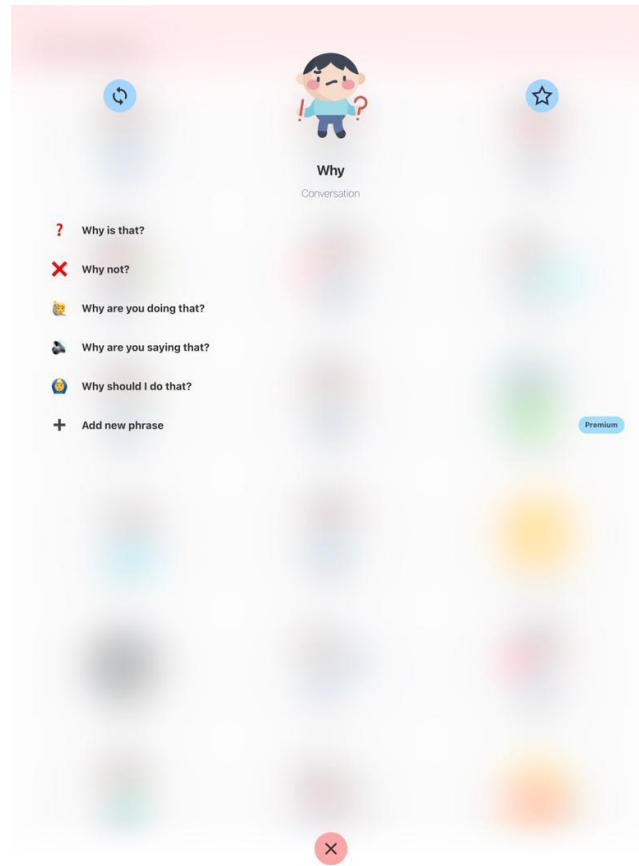


Figure 2.12: List of phrases on Leeloo when choosing Why as main topic

When users press the buttons and choose their titles, the phrases for the chosen titles will be listed for users to use as shown in Figure 2.12. It provides a number of pre-built phrases that users can use, but any custom phrases will only be available for premium users, which means that it is a paid feature. This restricts the customizability of the application, which also reduces its practicality as a long-term AAC solution for individuals who need highly personalized vocabulary.

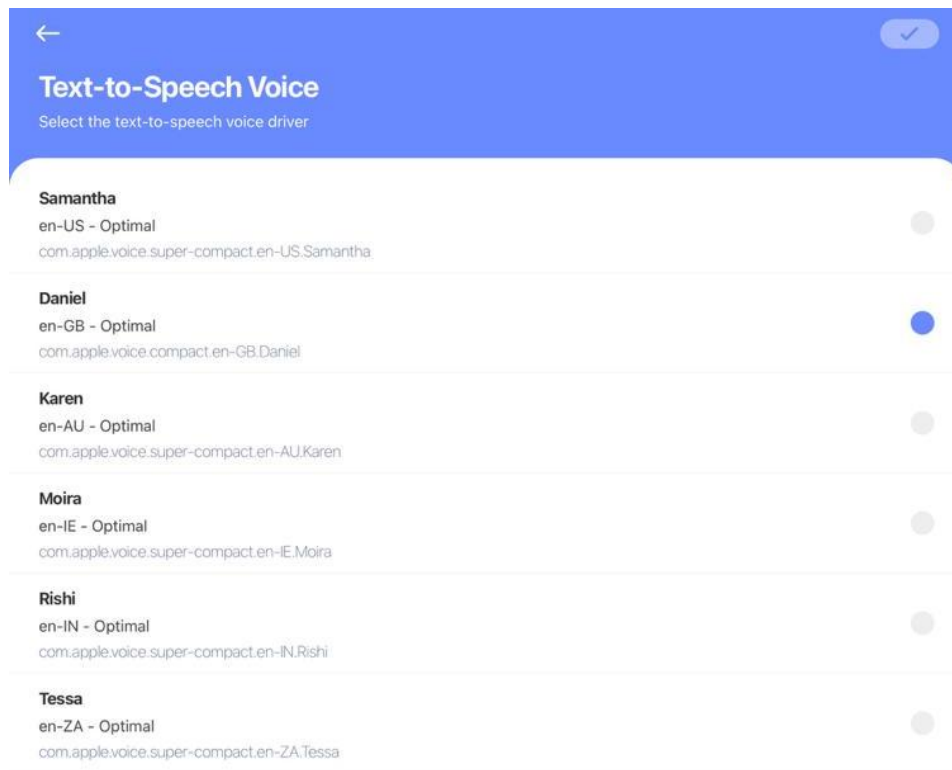


Figure 2.13: List of text-to-speech options for different voices on Leeloo

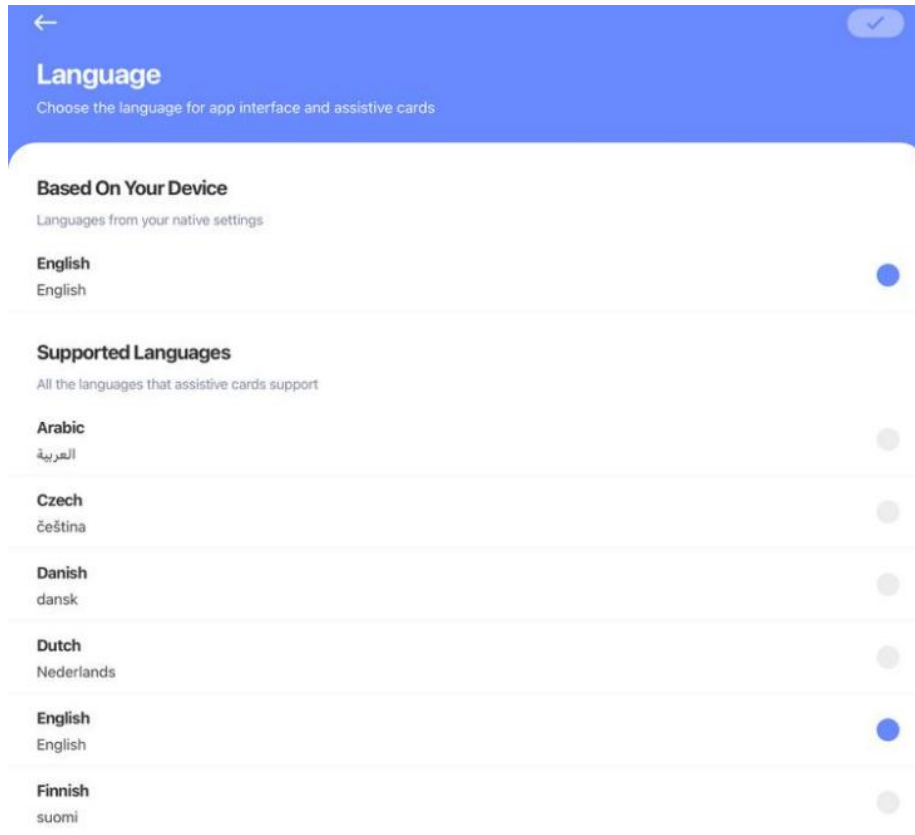


Figure 2.14: List of different languages that can be chosen on Leeloo

Text-in-speech function is also available on Leeloo. It provides a variety of different voices and accents, which can be chosen according to the user's preferences, as shown in Figure 2.13. Other than that, Leeloo also supports multi language where users are able to change their preferred languages for the app interface and the cards. This not only allow wider range of users to be able to use the app, but it also provides a decent amount of customizability when it comes to choosing a preferred voice.

2.4.3 Strengths and Weaknesses

Table 2.3: Strengths and Weaknesses for Leeloo

Strengths	Weaknesses
Very child-friendly UI design	Too child-centric, not suitable for adults

Text-to-speech function available	Vocabulary will be limited in the free version
Wide language support	Not ideal for complex sentences

One main advantage of Leeloo is that its design is very child-oriented, making it easier for non-verbal children to be able to use the application, as stated in Table 2.3. Leeloo's interface uses large buttons, pastel colours, and clear icons, which allow young children and beginning communicators to navigate through the application independently.

But, as mentioned in table 2.3, a too child-centric design can be discouraging for autistic teens and adults to use. The design can risks infantilising autistic teenagers and adults, which are already often being preceived as eternal children by others. Designing with only young children in mind overlooks the fact that autistic children grow into autistic adults, who continue to need robust, age-appropriate communication support.

Other than that, as mentioned in Table 2.3, Leeloo also supports the Text-to-speech feature, where chosen phrases will be read aloud. This would undoubtedly allow beginners to communicate complete messages without having to construct sentences word by word, which will increase the communication speed.

2.5 System Comparison

Table 2.4: System comparison

Feature / System	Proloquo	PECSTalk lite	Leeloo	Proposed System (PicExchange)
Text-to-speech	✓	✓	✓	✓

Preserves Traditional PECS binders methodology		✓		✓
Allow customisability	✓		✓	✓

Table 2.4 shows that all of the reviewed applications include a text-to-speech function, which is a fundamental requirement for PECS-inspired AAC systems because it allows picture selections to be converted into audible speech output. This feature enables non-verbal users to participate more fully in everyday interactions, as their selections can be clearly heard by communication partners in classrooms, therapy sessions, and home environments.

Table 2.4 also shows that only PECSTalk Lite and the proposed PicExchange system explicitly preserve the traditional PECS binder methodology by mirroring tabbed pages and a dedicated sentence strip. This design is particularly beneficial for caregivers and educators already trained in PECS, as it reduces the learning curve and supports a direct transition from physical communication books to a digital format. In contrast, Proloquo and Leeloo are PECS-influenced but rely more on grid-style or category-based layouts, which provide greater linguistic flexibility but move away from the classic binder structure.

Furthermore, Table 2.4 highlights that all four systems support some degree of customisability, but the scope and ease of customisation differ across applications. Proloquo offers extensive options, such as editing grid layouts, adding personal photos, and selecting from many high-quality voices, making it suitable for highly individualised communication setups. Leeloo and PECSTalk Lite allow users to add their own pictures and modify cards or pages, while the proposed PicExchange system aims to build on this by providing intuitive drag-and-drop sentence construction, flexible category creation, and straightforward image upload so that caregivers can quickly align the vocabulary with the learner's local context and daily routines.

2.6 Conclusion

Table 2.1 shows that Proloquo offers a strong built-in picture library, clear onboarding for new users, and high-quality text-to-speech voices, but its visual controls and customisability remain constrained in the trial version and continued use depends on subscription access. Table 2.2 indicates that PECSTalk Lite most closely preserves the traditional PECS binder structure and sentence strip workflow while providing multilingual text-to-speech and a substantial symbol library; however, its 7-day trial, reduced capacity, and limited initial guidance make it unsuitable as a long-term standalone AAC solution. Table 2.3 highlights that Leeloo delivers a very child-friendly interface, text-to-speech, and wide language support, yet its highly child-centric design, restricted free vocabulary, and limited support for complex sentence construction reduce its suitability for older users and advanced communicators. Finally, Table 2.4 compares core features across all systems and shows that while every reviewed application includes text-to-speech and some customisability, only PECSTalk Lite and the proposed PicExchange system retain the PECS binder methodology, positioning PicExchange to combine this familiar structure with more flexible, age-appropriate, and locally customisable features.

CHAPTER 3

Methodology

3.1 Introduction

This project will be implemented as:

- A redesigned responsive front-end website for Wings Melaka, focused on information presentation and usability improvements for visitors and staff editors.
- A basic native Android PECS communication application (PicExchange) with image-based sentence construction and text-to-speech, running entirely on-device.

The website rework will use a modern HTML/CSS/JavaScript stack with a responsive layout to support multiple screen sizes and improved navigation, without adding server-side logic beyond what already exists on the hosting environment. The PicExchange application will be built using Android Studio as the official IDE for Android, with Kotlin (and minimal Java only if necessary) and XML layouts, and will rely solely on on-device storage (e.g. internal storage or a simple local database).

3.2 SDLC

The prototyping model is chosen as the iterative SDLC approach because both the website and PECS app require intensive usability feedback from non-technical users (public visitors, staff, and autistic children/caregivers) to refine layout, navigation, and interaction patterns. The same prototyping phases apply to both sub-projects, but the concrete work in each phase differs slightly between website and app.

Phase 1: Requirements Gathering and Analysis

For the website rework:

- Existing Wings Melaka web pages and navigation structure are reviewed to identify

usability issues, inconsistent layouts, font usage, and color schemes.

- Informal discussions with Wings staff clarify content update workflows, priority information for visitors (program details, contact info, donation channels), and constraints such as existing hosting and CMS (if any).

For PicExchange:

- The functional scope is restricted to four core features: adding custom images and sentence starters, organizing them into categories, drag-and-drop composition of sentence strips, and basic text-to-speech output using Android's built-in TTS engine.
- Several existing PECS/AAC apps with drag-and-drop sentence strips (e.g. PECS IV+ style) are studied to identify essential interaction patterns (tabs/pages, drag vs tap, sentence strip behaviour) and to ensure the planned feature set remains "basic" without multi-language profiles, cloud sync, or advanced data tracking.
- Requirements from teachers and caregivers focus on simplicity, large touch targets, minimal steps per task, and offline reliability in classroom/home environments.

Phase 2: Quick Design

For the website:

- Low-fidelity wireframes are produced for key pages (home, about, programmes, donation/information, contact) with an emphasis on:
 - Clear menu hierarchy that separates information for parents, donors, and volunteers.
 - Consistent typography, improved color contrast, and flexible grid-based layouts for desktop, tablet, and mobile.
- A simple sitemap and navigation flow are drafted to reduce click depth and avoid duplicated or hidden content sections.

For PicExchange:

- Sketches and simple mockups show the main PECS board screen, category selection views, the sentence strip area, and the "Add image/sentence starter" screens.
- UI flows are drafted for: adding a custom picture from gallery/camera, creating a category, dragging a picture to the sentence strip, and triggering TTS playback.
- Basic local storage structure is planned (e.g. folders or a simple SQLite schema) to store images, categories, and sentence starters while staying fully on-device and offline.

Phase 3: Build a Prototype

For the website:

- An HTML/CSS/JavaScript prototype is built implementing:
 - The new menu structure and page layouts, using responsive design (media queries / responsive grid or lightweight CSS framework).
 - Placeholder content blocks to test visual hierarchy, font sizes, and spacing.
- No server-side logic, login forms, or payment integrations are added, in line with the out-of-scope constraints.

For PicExchange:

- An Android prototype is created in Android Studio using Kotlin and XML layouts, implementing:
 - A basic main screen with categories grid and a sentence strip area using touch-friendly views and simple drag-and-drop based on Android's view drag-and-drop APIs.
 - A minimal "Add image" flow that allows picking an image and saving its label and category locally.
 - Basic text-to-speech playback using the Android TextToSpeech API, reading out the sequence of items currently on the sentence strip, using the device's installed TTS engine.
- Local, on-device storage is implemented either through a simple SQLite database or structured files, ensuring no dependency on online services in keeping with the defined scope.

Phase 4: Initial User Evaluation

For the website:

- Staff and a small sample of parents or visitors are asked to perform tasks in the prototype (e.g. "find program information", "find how to contact Wings", "find donation info") while observing time taken, navigation errors, and confusion points.
- Feedback is collected on readability, content grouping, color choices, and mobile viewing experience.

For PicExchange:

- Caregivers, teachers, and if possible, autistic children are observed using the prototype to

carry out simple communication tasks (e.g. “I want drink”, “I want toy”), using the available images and sentence strip.

- Observations focus on how easily users understand drag-and-drop, whether buttons and text are large and clear enough, and whether TTS timing and volume feel appropriate for classroom/home settings.

Phase 5: Refining the Prototype

For the website:

- Navigation labels, menu order, and page layouts are refined based on user feedback to reduce confusion and improve discoverability of key content.
- Typography, spacing, and color contrasts are adjusted to meet basic readability and accessibility guidelines (e.g. larger headings, more whitespace, better contrast).

For PicExchange:

- Layout and interactions are tuned to simplify the number of steps needed to form a sentence, including larger icons, clearer feedback when dragging, and obvious “clear strip” controls.
- Error handling (e.g. accidental drag, deletion) is refined to prevent frustration for children, possibly by adding confirm prompts or undo for critical actions.
- No new out-of-scope features (multi-language, cloud sync, cross-platform) are added in this refinement; work remains focused on robustness and usability of the defined basic feature set.

Phase 6: Implementation and Maintenance

For the website:

- The refined front-end is prepared as production-ready HTML/CSS/JS and deployed to the existing Wings hosting environment, replacing the old front-end.
- A brief content update guide is prepared so staff can continue updating text and images with minimal technical knowledge.

For PicExchange:

- The prototype is hardened into a stable basic app, ensuring that local storage is reliable, images are properly managed, and TTS works across devices.
- An emulator (common Android screen sizes) is used to check layout consistency.
- Future maintenance is limited to bug fixes and small UI refinements within the existing

scope; advanced features are documented as potential future work, not implemented in this FYP.

3.3 Softwares

3.3.1 Android Studio

Android Studio is the official integrated development environment (IDE) for Android application development, provided by Google and built on the IntelliJ IDEA platform (Android Developers, 2026; Wikimedia Foundation, 2013). It bundles the Android SDK, project templates, code editor, emulator, and profiling tools in a single environment, which streamlines the process of designing, coding, testing, and packaging native Android applications (Android Developers, 2026; TechTarget, 2023).

One advantage of Android Studio is its strong support for user interface design through a visual layout editor, real-time previews on multiple screen sizes, and XML layout tooling, which allows developers to rapidly prototype and refine responsive interfaces for different Android devices (Android Developers, 2026; Intelivita, 2025). This is particularly useful for applications such as PicExchange that require frequent layout adjustments to optimize usability for children and caregivers.

Another advantage is the integrated development and debugging ecosystem, including the Android emulator, logcat, and performance profilers for memory, CPU, and network usage (TechTarget, 2023; Tencent Cloud, 2025). These tools help developers identify issues early, improve app performance, and ensure stability across a range of devices, which is important when the application will be used in real-world educational and home settings. Additionally, Android Studio provides first-class support for Kotlin and Java, code completion, and refactoring features, which increase development productivity and code quality (Android Developers, 2026; Intelivita, 2025).

3.3.2 Web Development Stack (HTML, CSS, JavaScript with Code Editor & Browser)

The Wings Melaka website redesign uses a standard front-end web stack consisting of HTML, CSS, and JavaScript as the core client-side technologies. HTML provides the structural markup

for content such as headings, paragraphs, images, and navigation menus, while CSS controls the presentation layer including layout, typography, colors, and spacing (GeeksforGeeks, 2018). JavaScript is used to add client-side interactivity where needed, such as responsive menus or simple dynamic behaviours, without introducing new backend functionality, which aligns with the project scope.

One advantage of using HTML and CSS is that they are universally supported open web standards, ensuring that the redesigned website can be accessed from a wide range of browsers and devices without additional plugins (GeeksforGeeks, 2018). CSS also enables responsive design through media queries and flexible grid layouts, which allows a single codebase to adapt to desktop, tablet, and mobile screens, thereby improving accessibility for different user groups (GeeksforGeeks, 2018).

Another advantage of this stack is that JavaScript can enhance user experience on the client side while keeping the architecture simple and maintainable. Combined with a modern code editor (such as Visual Studio Code) and built-in browser developer tools, developers can iteratively inspect elements, adjust layouts, and debug scripts directly in the browser, which supports the prototyping model used in this project. This approach is suitable for a UI-focused rework where the goal is to improve navigation, readability, and visual consistency without changing the underlying server or database.

3.4 Languages

3.4.1 Kotlin

Kotlin is a modern, statically typed programming language that has been officially supported by Google as a first-class language for Android development and is recommended as the preferred option for new Android projects (Android Developers, 2017; Lardinois, 2019). It runs on the Java Virtual Machine and is fully interoperable with existing Java code, allowing developers to gradually adopt Kotlin in Android applications (JetBrains, 2021).

One key advantage of Kotlin is its concise syntax, which reduces boilerplate code compared to Java and can make Android source code shorter, clearer, and easier to maintain (Android Developers, 2017; JetBrains, 2021). This is beneficial in an FYP context where development time is limited and code clarity is important for future maintenance and grading.

Another advantage is Kotlin's built-in null-safety features, which help prevent common runtime errors such as null pointer exceptions by enforcing nullability at the type system level (Android Developers, 2017; Lardinois, 2019). This improves reliability and reduces crashes, which is important for an assistive communication application used by children and caregivers. Combined with features like extension functions and coroutines, Kotlin supports writing more expressive and robust Android code while still leveraging the existing Java ecosystem (JetBrains, 2021).

3.5 Testing

The testing strategy for this FYP combines black box testing, light white box testing, and user-based usability testing to ensure both the redesigned Wings Melaka website and the PicExchange application function correctly and are practical for real users. Black box testing focuses on validating the observable behaviour of the system against the specified requirements, without considering the internal code structure, for example checking whether visitors can successfully navigate to programme information or whether users can create and speak a simple sentence using images on the sentence strip (GeeksforGeeks, 2018). Basic white box testing is applied to critical logic in the PicExchange app, such as storing and retrieving images, managing categories, and iterating through the sentence strip in the correct order, to verify that important code paths and conditions behave as expected (GeeksforGeeks, 2018). In addition, usability testing is conducted with representative users: public visitors and staff test the website to identify navigation, readability, and layout issues, while teachers, caregivers, and where appropriate autistic children interact with PicExchange to observe how easily they can add images, form sentences, and trigger text-to-speech output in realistic communication scenarios. Insights from these sessions guide iterative interface refinements to improve clarity, reduce task steps, and minimize user errors.

3.6 Target User

The redesigned Wings Melaka website primarily targets public visitors seeking information about the organisation's programmes and services, including parents of children with special needs, potential volunteers, and potential donors who require clear, up-to-date information and simple

navigation. Internal staff who update the website content are also key users, as they rely on a consistent layout and clear content structure that make it easier to maintain and expand pages over time without extensive technical knowledge. The PicExchange application targets autistic children and other non-verbal or minimally verbal users who benefit from picture-based communication using a simple PECS-style approach, where images and sentence starters are combined on a sentence strip to express basic needs and preferences (Pyramid Educational Consultants, n.d.). Caregivers, teachers, and therapists form an important secondary user group because they configure the categories, add custom images that reflect each child's environment, and support the child's daily use of the app in home or classroom contexts (Kids Club ABA, 2025). Both components of this FYP are therefore designed with accessibility, simplicity, and reliability in mind, so that they can be used effectively by individuals with varying levels of technical ability and communication needs.

3.7 Conclusion

This chapter has described the methodology for implementing the two components of this FYP: the front-end redesign of the Wings Melaka website and the basic Android PECS application, PicExchange. The prototyping model was selected as the guiding SDLC approach so that early versions of both the website and the app can be evaluated and refined based on feedback from real users, including public visitors, staff, caregivers, and autistic children (TeamHub, 2024). The technology choices were aligned carefully with the defined scope: HTML, CSS, and JavaScript with a modern code editor and browser tools for the website, and Android Studio with Kotlin and XML layouts for PicExchange, without introducing out-of-scope backend features such as authentication, online payment, multi-language management, or cloud synchronisation (Android Developers, 2017; Android Developers, 2026; GeeksforGeeks, 2018). A testing strategy combining black box, focused white box, and usability testing ensures that functionality, reliability, and user experience are all addressed within the constraints of a final year project. Together, these methodological decisions support the goal of improving information access through the website and providing a simple, practical communication medium for autistic children through the PicExchange application.

CHAPTER 4

Implementation Plan

4.1 Introduction

In this chapter, the project will detail on the implementation plan and its flow by using the case diagram and the activity diagram under UML diagram section. The report also details the Gannt chart for the first FYP phase, and second FYP phase including its timestamp by day. Other than that, the chapter will also detail the prototype for the website, and the application.

4.2 UML diagram

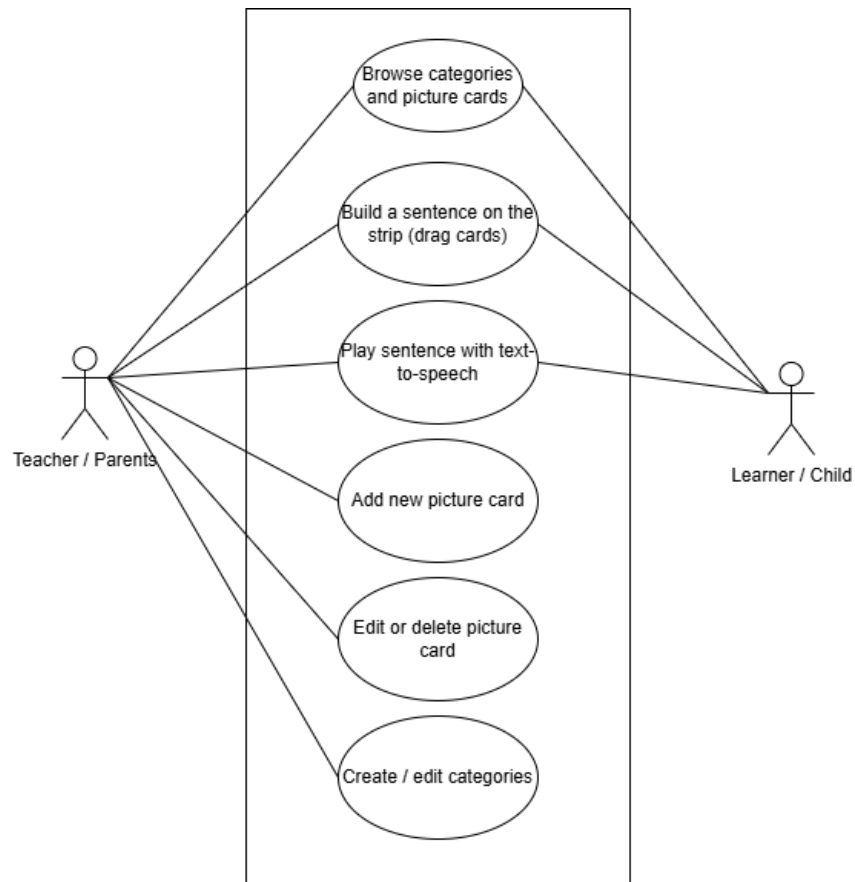


Figure 4.1: Use case diagram for PicExchange application

As shown in figure 4.1, the PicExchange PECS application has several use cases that are catered for 2 specific actors, which are Teacher or Parents, and Learner or Autistic children. As for Teacher/Parents, they are able to use all functions including Browsing the categories and pictures, they are also able to build a sentence using the pictures and sentence starter on the sentence strip, able to use text-to-speech function for pictures on the sentence strip, add custom pictures and sentence starter card, edit and delete picture cards and creating and deleting categories that were created. As for the Learner or Autistic children, they are limited to only browsing the categories and picture cards, building a sentence on the sentence strip using the picture cards and sentence starter, and using the text-to-speech function for pictures on the sentence strip.

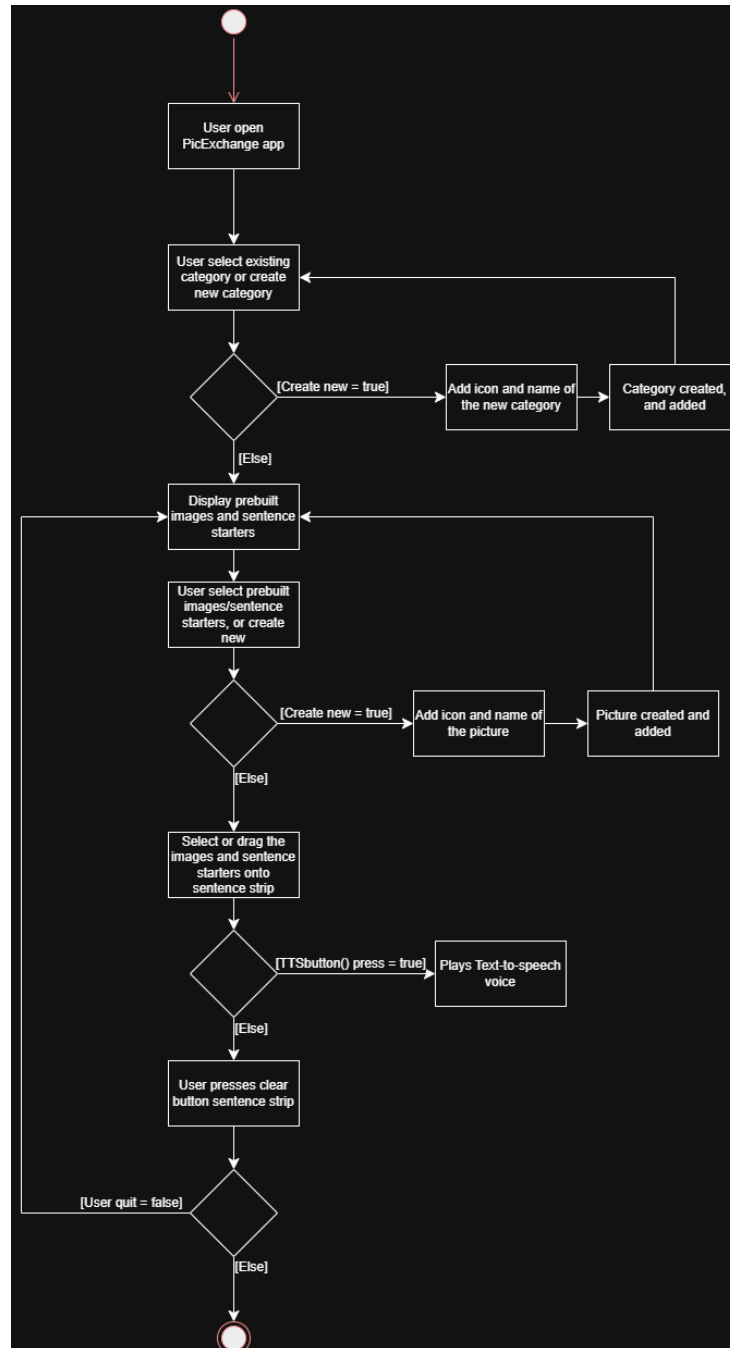


Figure 4.2: The activity diagram for the PicExchange PECS application

As shown in Figure 4.2, this is the flow of the PicExchange PECS application. User will firstly run the Application on any android device, and will be met with the list of categories for the pictures and sentence starters. At this stage, users can either choose to select among the existing categories, or create a new custom category. Should the user choose to create a new category, they

will be prompted to upload a picture or use a pre-built icon inside the application, and will be prompted to enter the name of the category. After that is done, user can agree with the changes and the new category will be created. If the user chooses to select the existing category, a library of pictures and sentence starters will be displayed according to chosen categories. At this point, users are able to choose the pictures and sentence starters and drag them onto the sentence strip, or creating a new picture. Should the user chooses the option to add a new picture, they will be prompted to upload a new picture, and will be prompted to name the picture. After that is done, the new picture will be created with the name user has typed in, with the picture user has uploaded. After users have dragged the pictures and sentence starters on to the sentence strip, users are able to use the text-to-speech function where the pictures and sentence starter that is currently on the sentence strip to be read aloud. Users are able to use the clear sentence strip function to remove all sentence starters and pictures on the sentence strip. The program will stop once users exited the program.

4.3 Gantt chart for phase 1 / 2

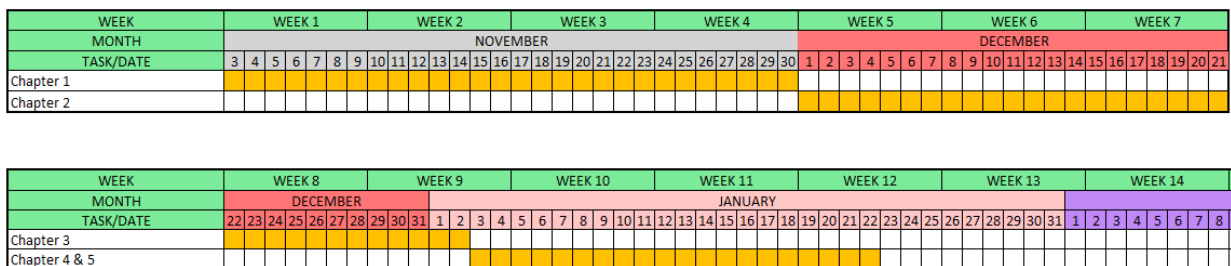


Figure 4.3: Gantt chart for phase 1 FYP

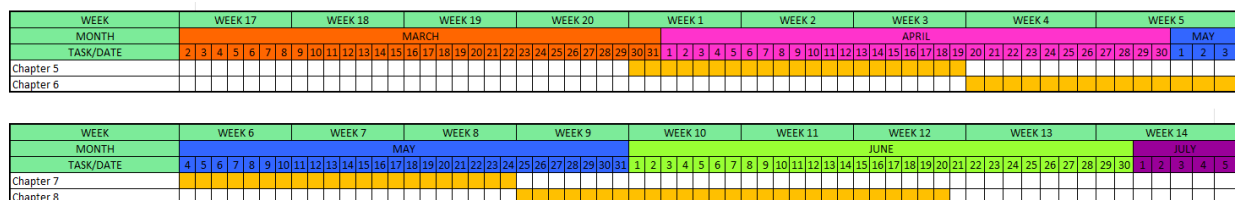


Figure 4.4: Gantt chart for phase 2 FYP

4.4 Design prototype

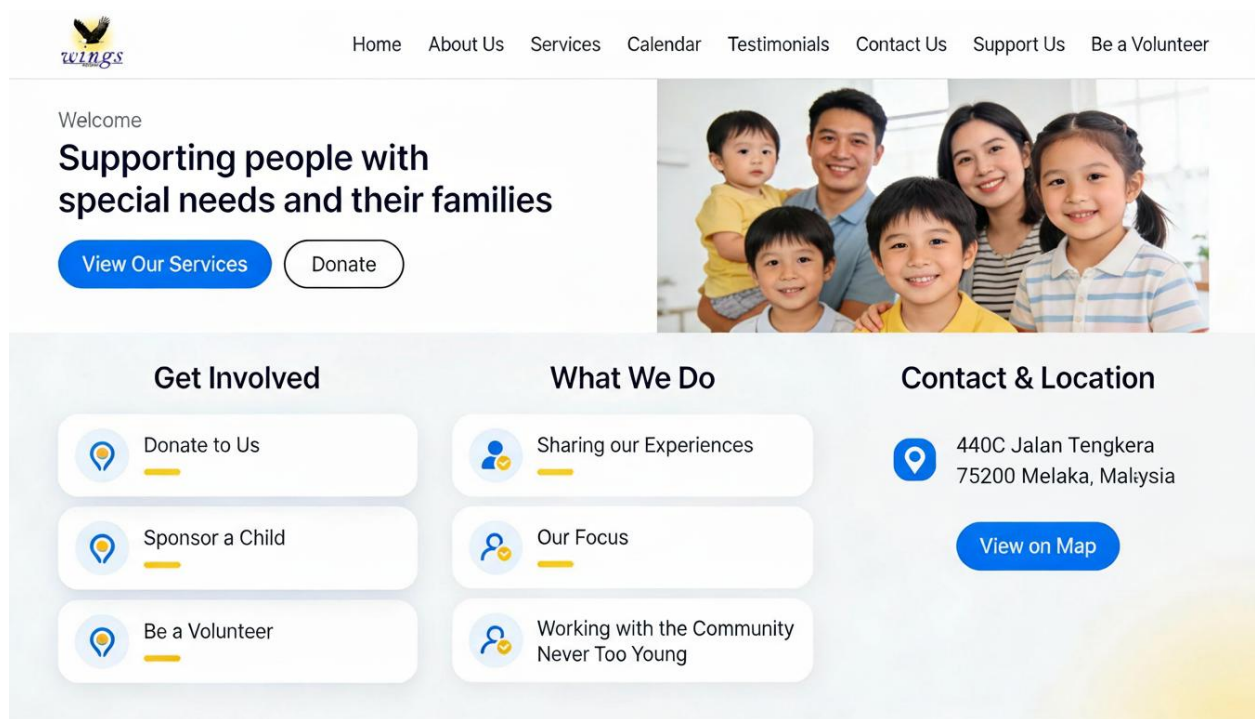


Figure 4.5: Prototype of rework design for Wings Melaka website.



Figure 4.6: Prototype of the homepage for PicExchange

CHAPTER 5

Conclusion

The report concludes that the FYP successfully lays a solid foundation for both improving Wings Melaka's online presence and providing a practical AAC tool for autistic learners.

The project addresses two core needs at Wings Melaka: clearer, more accessible information for the public through a redesigned front-end website, and a PECS-style Android application (PicExchange) that supports non-verbal or minimally verbal users in expressing their basic needs using image-based sentence construction with text-to-speech. The chosen scopes intentionally avoid complex backend or cloud features so that both deliverables remain realistic for a final year project yet still meaningful for real users in home and classroom environments.

The literature review on existing AAC/PECS-related apps (Proloquo, PECSTalk Lite, Leeloo) highlights that while current systems provide strong text-to-speech and symbol libraries, they often suffer from subscription lock-in, limited trial functionality, child-centric designs, or reduced customisability, especially in free versions. PicExchange is positioned to leverage the strengths of these systems—such as PECS-style sentence strips and flexible categories—while avoiding major weaknesses by emphasising offline use, basic but meaningful customisation, and a more age-appropriate, adaptable interface.

Using a prototyping SDLC model, with repeated cycles of quick design, prototype building, and user-focused evaluation, ensures that both the website and the PicExchange app can be iteratively refined around usability feedback from staff, caregivers, and autistic children. The selected technologies—HTML/CSS/JavaScript for the website and Android Studio with Kotlin and XML for PicExchange—support responsive layouts, touch-friendly interfaces, and stable on-device operation, matching the project's constraints and user requirements.

For the website, the redesigned navigation, responsive layout, and clearer content structure are expected to help parents, donors, volunteers, and staff access and manage information more easily, improving Wings Melaka's visibility and communication with stakeholders. For PicExchange, features such as category-based picture organisation, drag-and-drop sentence strips, and text-to-speech output provide a low-cost, portable alternative to traditional PECS binders, potentially reducing material maintenance while supporting more consistent communication across school and home.

Overall, the FYP demonstrates that a focused combination of front-end web redesign and a basic Android PECS application can meaningfully support Wings Melaka's dual goals of improving public information access and enhancing everyday communication for autistic children, while leaving more advanced capabilities such as multi-language management, cloud synchronisation, and cross-platform support as clear directions for future expansion beyond the current project scope.

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Appendix



Faculty of Information Science and Technology (FIST)
Final Year Project Meeting Log

MEETING DATE: 5/12/2025	MEETING NO.: 1
PROJECT ID: T87M743	
PROJECT TITLE : WINGS MELAKA: WEBSITE REVAMP/MAINTENANCE &AMP; PECS APP BUILDING	
TRIMESTER : 1	SUPERVISOR : ROY CHANG KWANG YANG
STUDENT ID & Name: 1221305591 MUHAMMAD FAISYMAN HAIQAL BIN MOHD FADZLI	CO- SUPERVISOR :

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.] Chapter 2
2. WORK TO BE DONE Improve Chapter 2
3. PROBLEMS ENCOUNTERED None
4. COMMENTS

Supervisor's Signature &
Stamp

Co-Supervisor's Signature
& Stamp (if any)

Student's Signature

NOTES:

- Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
- For FYP Phase 1, total six log sheets are to be submitted (every other week*).
- For FYP Phase 2, total six log sheets are to be submitted (every other week**).
- Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

**.: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)



Faculty of Information Science and Technology (FIST)
Final Year Project Meeting Log

MEETING DATE: 26/12/2025	MEETING NO.: 2
PROJECT ID: T87M743	
PROJECT TITLE : WINGS MELAKA: WEBSITE REVAMP/MAINTENANCE & PECS APP BUILDING	
TRIMESTER : 1	SUPERVISOR : ROY CHANG KWANG YANG
STUDENT ID & Name: 1221305591 MUHAMMAD FAISYMAN HAIQAL BIN MOHD FADZLI	CO- SUPERVISOR :

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.] Chapter 2
2. WORK TO BE DONE Chapter 1, 3 and 4
3. PROBLEMS ENCOUNTERED None
4. COMMENTS

Supervisor's Signature &
Stamp

Co-Supervisor's Signature
& Stamp (if any)

Student's Signature

NOTES:

5. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
6. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
7. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
8. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

**.: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)



Faculty of Information Science and Technology (FIST)
Final Year Project Meeting Log

MEETING DATE: 2/1/2026	MEETING NO.: 3
PROJECT ID: T87M743	
PROJECT TITLE : WINGS MELAKA: WEBSITE REVAMP/MAINTENANCE &AMP; PECS APP BUILDING	
TRIMESTER : 1	SUPERVISOR : ROY CHANG KWANG YANG
STUDENT ID & Name: 1221305591 MUHAMMAD FAISYMAN HAIQAL BIN MOHD FADZLI	CO- SUPERVISOR :

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.] Chapter 1
2. WORK TO BE DONE Chapter 3 and 4
3. PROBLEMS ENCOUNTERED None
4. COMMENTS

Supervisor's Signature &
Stamp

Co-Supervisor's Signature
& Stamp (if any)

Student's Signature

NOTES:

9. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
10. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
11. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
12. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

**.: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)



Faculty of Information Science and Technology (FIST)
Final Year Project Meeting Log

MEETING DATE: 9/1/2026	MEETING NO.: 4
PROJECT ID: T87M743	
PROJECT TITLE : WINGS MELAKA: WEBSITE REVAMP/MAINTENANCE &AMP; PECS APP BUILDING	
TRIMESTER : 1	SUPERVISOR : ROY CHANG KWANG YANG
STUDENT ID & Name: 1221305591 MUHAMMAD FAISYMAN HAIQAL BIN MOHD FADZLI	CO- SUPERVISOR :

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.] Chapter 3
2. WORK TO BE DONE Chapter 4 and 5
3. PROBLEMS ENCOUNTERED None
4. COMMENTS

Supervisor's Signature &
Stamp

Co-Supervisor's Signature
& Stamp (if any)

Student's Signature

NOTES:

13. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
14. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
15. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
16. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

**.: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)



Faculty of Information Science and Technology (FIST)
Final Year Project Meeting Log

MEETING DATE: 16/1/2026	MEETING NO.: 5
PROJECT ID: T87M743	
PROJECT TITLE : WINGS MELAKA: WEBSITE REVAMP/MAINTENANCE & PECS APP BUILDING	
TRIMESTER : 1	SUPERVISOR : ROY CHANG KWANG YANG
STUDENT ID & Name: 1221305591 MUHAMMAD FAISYMAN HAIQAL BIN MOHD FADZLI	CO- SUPERVISOR :

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.] Chapter 4
2. WORK TO BE DONE Chapter 5
3. PROBLEMS ENCOUNTERED None
4. COMMENTS

Supervisor's Signature &
Stamp

Co-Supervisor's Signature
& Stamp (if any)

Student's Signature

NOTES:

17. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
18. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
19. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
20. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

**.: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)

FYP for FIST: Attendance List

Session 4: 12/12/2025

The following students have attended the FYP session (FIST) with the


MELISSA DE VALDA BINTI MOHD YATIM
Assistant Manager, Library
Information Skill & Research Support Unit
Multimedia University
Jalan Ayer Keroh Lama
75450 Melaka

library. Trainer:

Melissa De Valda Binti Mohd Yatim
Librarian, Tun Dr. Hasmah Mohd Ali Digital Library

No.	Full Name	MMU ID No.	FYP title
1	Soo Kian Shenh	1211106761	Resource-aware Federated Learning for Temporary Enlisted Heterogeneous Compute Node
2	Leong Ka Hao	1231303580	Deepfake Detection
3	LIM CHIA QUAN	243UT246X1	MARKET BASKET ANALYSIS TO ENHANCE SALES STRATEGIES IN LOCAL RETAIL BUSINESS
4	LEW ZIXIN	243UT246WZ	SMART TRAVEL PLANNING MOBILE APP WITH DYNAMIC ITINERARY OPTIMIZATION
5	Rishma Devi A/P Sivaraj	1221101095	Event Visualize Dashboard
6	Kuan Ing Shien	1231302893	AI Waste Sorting Assistant
7	Muhammad Sahil Qayyum Bin Mohd Yahya	1211112300	Advancing Trust Assurance in Vehicular Ad Hoc Networks
8	TANZEKANG	1221101804	XAI stress factor
9	TAN JUN YAN	1211109993	STOP SCHOOL BULLY REPORTING APP
10	Fang Yong Aun	1231302733	Item Defect Detection System
11	U BOON XUAN	1211110723	Genomics analysis of meningitis-causing bacteria
12	Tan Joon Shen	1211108635	MathViz
13	Tee Xue Ni	243UT246WB	Securing drones from GPS Spoofing attacks
14	Dylan Ang Zi Jun	1211110281	Image Waste Classification Using machine learning approach
15	Beckham Wong Yao Kang	1211107299	3DAGENT: An Intelligent Conversational Agent in Matterport 3D Spaces
16	Akmal Muqry Bin Ainuddin	243UT246WP	IOT in Aquaculture
17	Ng Chze Yie	243UT246X3	Cloud-based IOT Health Monitoring System Using Raspberry Pi
18	Abhiraami A/P Sivakumar	1211111516	AI-Enhanced Medical Imaging For Diagnosis
19	MUHAMMAD AIMAN ISKANDAR	1211107975	FACE AND LIVENESS DETECTION BASED BANK LOCKER

20	ANGIE EE EN QI	1211109081	GAMIFICATION PROGRAM USING AUGMENTED REALITY(AR)
21	Wong chuen khoon	243UT246W6	PERFORMANCE COMPARISON OF RPL LITE VS RPL CLASSIC IN 6LoWPAN IOT NETWORKS
22	Shi Hong Yi	1211108962	Motion Music Maestro: CV Body Movement to Orchestral Composition
23	Bryan Chin Yew Chung	241UT2416C	Multi-CCTV Continuity: Person Re-Identification and Cross Camera Tracking
24	Jayme Ang Zi Feng	1211110224	Brain Tumor Detection using machine learning
25	WANG XIN LING	1221102572	Smart Parking Availability Detection System Using Machine Learning Approaches
26	ESTHER LIM YI SHUEN	1211111787	User Experience Dashboard: Cybersecurity Awareness and Behaviour Analysis
27	Yap Ji Yun	242UT244BE	ZERO TRUST FRAMEWORK FOR SMES IN CLOUD ENVIRONMENTS
28	AMANDA LIM JIA XIN	1221102888	Smart Attendance System Using Face Recognition For Improved Authenticity

29	TENG YAO WEN	242UT2448Z	Feed Social System
30	LIM JUN HONG	1211110418	Blockchain-based vehicle service and warranty management system
31	Tanishaa Nair Kamalesvaran	1231302585	HEALTHCARE STAFF CYBERSECURITY AWARENESS TRAINING PLATFORM
32	KOH JUN KET	242UT24494	Vehicle breakdown and sos
33	Muhammad Anas Syazwi Bin Warman	1221305773	Federated Learning On Smartphone-Based Human Activity Recognition
34	NG WEN HUI	1221101649	IOT -BASED AUTOMATED FEEDER FOR PET
35	Zavier Ong Yi Chang	242UT243V9	ONLINE MOBILE GAS BOOKING SYSTEM
36	LIM WEI JIAN	243UT246WJ	TRAFFIC IMPACT ASSESSMENT REPORT GENERATION USING LLAMA
37	Muhammad Aswad Suhail Bin Mohamad Yasin	242UT241GT	DATA-DRIVEN ANALYSIS OF GAMEPLAY HOURS AND PLAYER SATISFACTION IN MODERN GAMES
38	LIM JUN YONG	243UT246XA	PERFORMANCE COMPARISON OF TSCH VS CSMA IN 6LoWPAN IoT NETWORKS
39	MUHAMMAD HAZIQ DANIEL BIN MOHD ZAMNI	1231302561	An Analysis of Financial Management Practices Among University Students
40	CHONG JIA SHENG	1231300976	TASKBEAM: A SMART TO-DO LIST APP
41	SAMUEL ABELLIN BAK JING ZHI	1211102332	T88J225 : ANALYSIS OF E-WALLET USAGE AND SPENDING TRENDS AMONG TERTIARY STUDENTS
42	Too Jun Cai	1211110886	Speech-to-text transcripator based on the Nvidia quartznet model
43	SUMI LEE YAN XIN	1221309444	Evolutionary Analysis of Genome Reduction in Pathogens
44	MEGAT NUR HASANAL PUTERA BIN NORRASIDI	1211112299	IMPLEMENTATION OF DEEP LEARNING FOR TEXT CLASSIFICATION ON CUSTOMER'S REVIEW
45	HASVINJIT KAUR	1221309285	ANALYSING THE IMPACT OF SOCIAL MEDIA USAGE ON PRIVACY AWARENESS AND DATA SHARING
46	Arumugam Kayman Srinivasan	1231301161	Earthquake Prediction using Deep Learning

47	Ethan Lim Zhi Xian	1211103291	In silico and ADMET Profiling of Phytochemicals from Malaysian Zingiberaceae
48	Mohnish Nanthakumar	1211111388	Secure file encryption and decryption system using AES with integrity verification
49	Si sheng fu	242UT24498	Development of smart clinic system
50	Yip Jia Xuan	1221309282	Development of an AI-Based Model for Pollen Viability Testing and Imaging Optimization
51	MUHAMMAD FAISYMAN HAIQAL BIN MOHD FADZLI	1221305591	WINGS MELAKA: WEBSITE REVAMP/MAINTENANCE & PECS APP BUILDING
52	TAM YU SIONG	243UT246X5	AI POWERED FINANCIAL TRACKER
53	Charlene Tay Huey Wen	1231303594	Smart Queue Management System With Secure Network Communication
54	Mohan Raj A/L Vesunathan	1221308494	REAL-TIME IOT SENSOR NETWORK FOR AUTOMATED FIRE RESPONSE
55	Kelven Chew Khye Wee	1211111644	DURIAN FARM INFORMATION AND TRACEABILITY WEB APPLICATION
56	Dennis Wee Khor Siong	1211112008	Single-Sample Face Recognition for Biometric Security Applications
57	Muhammad Faris Aidel Bin Dahlan	1211112319	The Study of Bandwidth Recycling in 4G/5G
58	ONG YONG QUAN	243UT246XG	IDENTIFYING DEEPFAKE IMAGES
59	Michelle Theba A/P Jeffrey Paul Raj	1231302828	Dimension Reduction and Machine Learning for Multimodal Data
60	CHAI QI YANG	243UT246XE	LIVENESS DETECTION FOR MITIGATING DISPLAY REPLAY ATTACKS
61	Joshua Lai Jia Jian	1211102761	web-based wifi tracking system
62	LAU YONG PIN	243UT246XR	LANDMARK LOCALIZATION IN CNN-BASED MASKED FACE RECOGNITION
63	Junior Gregory Danker	1231302631	ENHANCING ENGAGEMENT IN ONLINE LEARNING THROUGH ADAPTIVE CONTENT DELIVERY
64	Chern Hao Wen	243UT246XN	DATA STORAGE SYSTEM BY PDF417
65	CYRUS RHYS RONNY	1231303085	Campus Network Monitoring and Intrusion Detection Simulation
66	Amira Tsara Bt Mohd Zamri	1231302647	Music Recommendation System Based on User's Facial Age and Gender Recognition
67	Bong Li Hen	243UT246WG	Lost and Found
68	LUKMANNULHAKIM BIN MUHAFIF	1211109429	FOREST FIRE ANALYSIS USING GOOGLE EARTH ENGINE (GEE)

69	Tang Wei Jie	1221102959	A WEB BASED FOOTBALL PREMIER LEAGUE SCORE PREDICTION SYSTEM
70	KOOH SHU JUN	243UT246WT	T88J129 IMPLEMENTATION OF A LATTICE-BASED IDENTIFICATION SCHEME FOR POST-QUANTUM AUTHENTICATION
71	Guna sureyah	123101245	Boosting based heart disease classification model
72	WONG KAI SIN	242UT2449C	A SECURE MOBILE FINANCE MANAGEMENT APPLICATION
73	Saw Wei Wen	242UT243V2	A study of neonatal baby reflex framework

74	Cheng Yu He	243UT246WU	DESIGN AND DEVELOPMENT OF A WUTHERING WAVES GAME GUIDE SYSTEM
75	Annusyah Youmesan	1211111761	AI-Driven Secure IoT Vehicle Tracking and Anti-Theft System with Voice Authentication
76	Low Ming Yuan	1211111801	AI-BASED TOURISM REVIEW ANALYSIS AND SUMMARIZATION SYSTEM USING DEEP LEARNING
77	THARANI A/P KUMARAN	1221102419	Comparison of Cross-Species Host Response to Candida Albicans
78	MAADHUMITHAA A/P BALAKUMARA	1221100837	Attendance management system using face recognition
79	Chia Jun Le	1231302612	Inventory and Sales Management Application for Small Business
80	TAN WEN JIAN	243UT246WH	Implementation of Virtual Reality (VR) Mobile App
81	Lim Liang Yea	242UT24492	LIBRARY INFORMATION AND AI-ASSISTED BOOK SEARCH WEB APPLICATION
82	Mah Kah Hei	242UT243V0	RECYCLE CITY: A CITY ENVIRONMENT CLEANING SIMULATOR GAME
83	Felix Chong Shin	1231303188	Implementing of a provably secure e-cheque scheme
84	Darren new jiun en	243UT246WF	Enterprise Chatbot using locally hosted open-weight large language model
85	Anas Ahmed Hassan Ahmed	1221103131	Implementation of a secure, responsive university web portal
86	Gareth Tan Yao Wei	1221305518	Event Planning with Real-Time Budget Monitoring System
87	YEW CHOON KIAT	242UT243V8	HANDOVER DELAY IN IPV6 IOT
88	MUHAMMAD RAFIQI BIN RUSHDAN	1231302251	BAGGING-BASED HEART DISEASE CLASSIFICATION MODEL
89	Arif Aiman bin Azman Hoh	1231301449	T-Pot Honey-pot Analysis with MITRE ATT&CK Framework
90	Leong Wung Jie	1211109377	IOT PARALYSIS PATIENT HEALTH CARE PROJECT
91	Muhammad Aswad Suhail Bin Mohamad Yasin	242UT241GT	DATA-DRIVEN ANALYSIS OF GAMEPLAY HOURS AND PLAYER SATISFACTION IN MODERN GAMES
92	Chong Ming Chai	243UT246XB	AI-StudyMate
93	Amirul haqimi bin suhaimi	1221305594	AUTONOMOUS MEDICAL DELIVERY ROBOT
94	GOH WEE YANG	243UT246WE	INTERNET OF TOYS
95	ONG CHANG YONG	1211108837	CampusCycle
96	Looi Guang Yong	242UT2440N	A study exploring the Internet of Drones
97	KOH JIN RONG	1211109043	A CONVERSATIONAL AI-POWERED FINANCIAL ASSISTANT WITHIN CHAT PLATFORMS
98	TEY ZHI HONG	1211108713	REVIEW AND ANALYSIS OF POST-QUANTUM IDENTIFICATION SCHEMES
99	YONG JING YU	1221304522	AI-POWERED APPLICATION FOR AUTOMATED PRODUCT COUNTING
100	Loh Jie Hao	1211108262	T89O481 : INTERACTIVE UNIVERSITY EVENT SHOWCASE AND REGISTRATION PLATFORM
101	Hon Xue Yong	1221305578	IoT-enabled sustainable urban mobility with edge-powered incentive schemes

102	TAI KE YING DOROTHY	1211111348	Smart Inventory Management System using rfid and lot
103	TAN ZI JIAN	1161202476	BADMINTON BOOKING AND MANAGEMENT SYSTEM
104	WONG KAR HOE	1211108111	ABNORMAL CROWD BEHAVIOUR DETECTION USING MACHINE LEARNING
105	DHURKAA BALA KRISHNAN	1221101117	COMPARATIVE GENOMIC ANALYSIS OF OXALOTROPHIC BACTERIA LINKED TO UROLITHIASIS
106	DANIEL TAN NAI LE	242UT24495	CLOUD BASED WEB APPLICATION
107	SOFIA BATRISYIA BINTI MOHAMAD FARIS	1211111880	Deep Learning Model Analysis for Network Anomaly Detection
108	Anis Syifaa' binti Mohd Zaffarin	1211112369	AI-Assisted Writing System for Dyslexic Students

109	DARWISYA ARINA BINTI ZAHAMI	1211112325	DEEP LEARNING TO DETERMINE HEALTH OUTCOMES FOR MENTAL DISORDERS USING EHR
110	Ng Hai Zhu	1231303122	AUTOMATED CAMPUS GATE ACCESS SYSTEM
111	NUR INSYIRAH IMAN BINTI MOHD AZMAN	1211112312	AUTOMATED DETECTION OF PARKINSON'S DISEASE FROM GAIT PATTERNS USING AI
112	CHONG KAI WAI	242UT2449N	ROOM AND FACILITY MANAGEMENT SYSTEM
113	Saidatul Huda Binti Mohd Hasbullah	1231301491	Design and Implementation Of a Cloud Infrastructure Using All-In-One Openstack
114	Luqman Hakim Bin Mohamad Zaki	1231303290	Comparative Study of MobileNet and EfficientNet for Optimized Crop Disease Detection

